

Mid-term examination on
Logic Circuit Design
(Group 1)

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17 October 2007

1 Balanced Ternary corrected

Question. Find a formula with a fixed number of operations for

$$S(q, n) = q^{n-1} + q^{n-2} + \cdots + q + 1, \quad \text{where } 1 < q.$$

What if $q = 2$ and $q = 3$?

Answer. We have

$$S(q, n) = q^{n-1} + q^{n-2} + \cdots + q + 1 \tag{1}$$

$$q \times S(q, n) = q^n + q^{n-1} + \cdots + q^2 + q \tag{2}$$

Let us subtract (2) from (1):

$$\begin{aligned} qS(q, n) - S(q, n) &= q^n - 1 \\ (q - 1)S(q, n) &= q^n - 1 \end{aligned}$$

Since $q \neq 1$, we have

$$S(q, n) = (q^n - 1)/(q - 1)$$

Therefore $S(2, n) = 2^n - 1$ and $S(3, n) = (3^n - 1)/2$.

Question. Prove that if the leftmost nonzero trit is 1, then the number is positive, otherwise it is negative.

Answer. Let us consider a number N whose n -trits representation is

$$1t_{n-2}t_{n-3}\dots t_0$$

in balanced ternary ($t_i \in \{\bar{1}, 0, 1\}$). By definition, this means that $N = 3^{n-1} + t_{n-2}3^{n-2} + \dots + t_0$. In order to prove that $0 < N$, let us take the lowest possible value of N and show it is indeed positive. This value N_{min} is defined by taking all the remaining unspecified trits to be $\bar{1}$, that is

$$\begin{aligned} N_{min} &= 3^{n-1} - 3^{n-2} - \dots - 1 \\ &= 3^{n-1} - (3^{n-2} + \dots + 1) \\ &= 3^{n-1} - S(3, n-1) \\ &= 3^{n-1} - (3^{n-1} - 1)/2 \\ &= (3^{n-1} + 1)/2 \end{aligned}$$

So $0 < N_{min}$, and thus so is N .

Let us suppose now that N has a balanced ternary representation of the shape $\bar{1}t_{n-2}t_{n-3}\dots t_0$. By definition, this means that $N = -3^{n-1} + t_{n-2}3^{n-2} + \dots + t_0$. In order to prove that $N < 0$, let us take the greatest possible value of N and show it is indeed negative. This value N_{max} is defined by taking all the remaining unspecified trits to be 1, that is $N_{max} = \bar{1}11\dots 1 = \overline{111\dots 1} = \overline{N_{min}} = -N_{min} < 0$. (Complementing the trits is the same as negating the number.) So $N < 0$.

2 Ternary

Question. Write an **efficient** C function which converts from ternary to decimal. The prototype is

```
int from3to10 (int* ternary, int t_size);
```

where `ternary` is an array of integers, of size `t_size`, representing a number in base 3. The most significant digit (also known as *trit*) is the first element of array `ternary`.

Answer.

```
#include<stdio.h>

int from3to10(int* ternary, int t_size) {
    int decimal=ternary[0];
    for (int index=1; index<t_size; index++) {
        decimal = decimal * 3 + ternary[index];
    }
    return decimal;
}
```

```

}

int main() {
    int t [3] = {1,1,2};
    printf("%d\n",from3to10(t,3));
    return 0;
}

```

3 Conversions

Question. Decipher the secret message by converting **efficiently** from octal to hexadecimal the following numbers:

5413 3726355 12 157255 150320

Answers. One efficient way consists in converting each digit to 3-bit binary numbers, then grouping them four-by-four bits, from the right to the left, and finally converting separately the groups to hexadecimal:

$$\begin{aligned}
 5413_8 &= (5_8)(4_8)(1_8)(3_8) = (101_2)(100_2)(001_2)(011_2) \\
 &= (1011_2)(0000_2)(1011_2) = (B_{16})(0_{16})(B_{16}) \\
 &= B0B_{16} \\
 3726355_8 &= (3_8)(7_8)(2_8)(6_8)(3_8)(5_8)(5_8) \\
 &= (011_2)(111_2)(010_2)(110_2)(011_2)(101_2)(101_2) \\
 &= (1111_2)(1010_2)(1100_2)(1110_2)(1101_2) \\
 &= (F_{16})(A_{16})(C_{16})(E_{16})(D_{16}) \\
 &= FACED_{16} \\
 12_8 &= (1_8)(2_8) = (001_2)(010_2) = (1010_2) \\
 &= A_{16} \\
 157255_8 &= (1_8)(5_8)(7_8)(2_8)(5_8)(5_8) \\
 &= (001_2)(101_2)(111_2)(010_2)(101_2)(101_2) \\
 &= (1101_2)(1110_2)(1010_2)(1101_2) \\
 &= (D_{16})(E_{16})(A_{16})(D_{16}) \\
 &= DEAD_{16} \\
 150320_8 &= (1_8)(5_8)(0_8)(3_8)(2_8)(0_8) \\
 &= (001_2)(101_2)(000_2)(011_2)(010_2)(000_2) \\
 &= (1101_2)(0000_2)(1101_2)(0000_2) \\
 &= (D_{16})(0_{16})(D_{16})(0_{16}) \\
 &= D0D0_{16}
 \end{aligned}$$

Message: BOB FACED A DEAD DODO.